

Enhancing Retrosynthesis Prediction with Distillation Learning

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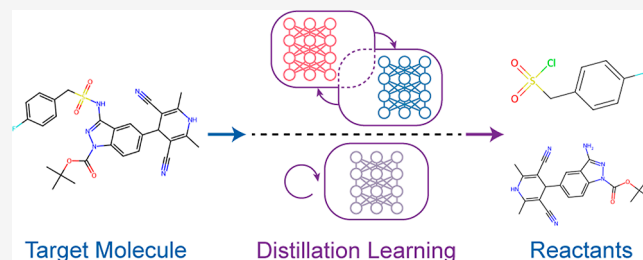


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ABSTRACT: Single-step retrosynthesis prediction aims to identify reactants for synthesizing a target molecule and is crucial for pathway planning. Despite improved prediction accuracy, existing methods often struggle with performance disparities between high- and low-resource reaction classes. This limitation hampers the overall effectiveness of the retrosynthesis prediction methods. Here, we introduce two novel strategies—Retrosynthetic Mutual Distillation (Retro-MD) and Retrosynthetic Self-Distillation (Retro-SD)—leveraging distillation learning techniques to bridge this performance gap. Retro-MD uses dual sampling temperatures and cross-model knowledge transfer, while Retro-SD leverages a fixed temperature and self-distillation from prior iterations. Evaluations on Transformer-based models show a state-of-the-art performance among template-free approaches. Ablation studies further validate the rationale of reaction-class-aware task partitioning, demonstrating the robust effectiveness of our proposed distillation strategies.



1. INTRODUCTION

Retrosynthesis is a fundamental aspect of organic chemistry, focusing on the identification of synthetic pathways that transform target molecules into commercially viable materials.^{1–3} This process not only enables the exploration of alternative synthetic routes to bypass patent restrictions but also facilitates the discovery of cost-effective synthesis methods.^{4–6} Notably, retrosynthetic strategies have driven advancements in drug design, establishing novel chemical domains within the pharmaceutical industry.^{7–10}

In recent years, despite the continuous emergence of new chemical synthesis technologies, the unknown reaction mechanisms and vast search space still pose significant challenges even for experienced chemists. As Artificial Intelligence (AI)-driven molecular generation models^{11,12} continue to produce novel molecules with desired properties, chemists also seek to predict synthetic pathways for these molecules through deep learning approaches.¹³ These deep learning-based synthesis methods have achieved accurate predictions of reaction kinetics,¹⁴ molecular chemical properties,^{15,16} reactivity,¹⁷ and synthesis planning,¹⁸ thereby driving the advanced development of retrosynthetic computational tools.^{19–21} This paper focuses on single-step retrosynthesis prediction in retrosynthetic planning, which involves determining the reactants required for the synthesis of a given product molecule.

The application of deep learning in chemistry has been particularly transformative in three key areas. First, generative modeling has revolutionized molecular design by enabling the creation of novel chemical structures with the desired

properties. Recent comprehensive reviews by Zeng et al.²² and Tang et al.²³ highlight how deep generative models, including variational autoencoders, generative adversarial networks, and diffusion models, are reshaping drug discovery by efficiently exploring vast chemical spaces. These approaches have demonstrated remarkable success in generating molecules with specific biological activities, pharmacokinetic properties, and synthetic accessibility. Second, machine learning approaches for chemical property prediction have evolved significantly, with sophisticated models capable of accurately predicting various molecular properties. Koutsoukos et al.²⁴ provided a comprehensive review of machine learning algorithms specifically applied to ionic liquid chemical space, while Fooladi et al.²⁵ conducted an extensive evaluation of over 11,000 machine learning models for molecular property prediction, emphasizing the critical importance of robustness on out-of-distribution data. Their findings revealed that graph neural networks generally exhibit superior robustness compared with classical machine learning approaches when facing distribution shifts. Third, the prediction of chemical reactivity has emerged as a crucial application area, with machine learning models demonstrating the ability to predict reaction outcomes, yields, and mechanisms. Zuranski et al.²⁶ reviewed

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the use of supervised learning for predicting reaction yields, while Raghavan et al.²⁷ provided guidelines for data set design in building models of chemical reactivity. Furthermore, Jorner et al.²⁸ comprehensively reviewed the evolution from traditional mechanistic approaches to machine learning methods in modeling organic reactivity and demonstrated in a related study²⁹ how combining mechanistic modeling with machine learning can accurately predict experimental activation energies with mean absolute errors as low as 0.77 kcal/mol. These advances collectively demonstrate the maturation of AI-driven approaches in chemistry and set the foundation for more sophisticated retrosynthesis prediction methods.

Existing single-step retrosynthesis prediction methods can be broadly categorized into template-based, semitemplate-based, and template-free approaches.³⁰ Template-based methods^{31,32} treat retrosynthesis prediction as a template-matching task. The seminal work by Segler and Waller pioneered the use of deep neural networks for template-based retrosynthesis, combining neural networks with symbolic AI to plan chemical syntheses efficiently.³³ Template-based methods rely on pre-extracted templates to construct a template library, encoding the core information on the reaction process, such as atom and bond transformations at the reaction center. These methods often focus on improving template matching algorithms and molecular representations. Among these solutions, GLN³² employs Conditional Graph Logic Networks to jointly model template matching and reactant graph matching probabilities. Similarly, LocalRetro³⁴ utilizes graph neural networks to identify local templates for specific atoms and bonds. While template-based methods offer notable advantages in efficiency and interpretability, their applicability is limited to molecules contained within the template library, resulting in poor generalization to novel reactions.

To overcome the limitations of template-based methods, researchers developed a semitemplate approach by mimicking the reasoning process of chemists. These methods decompose retrosynthesis into two stages: prediction of bond disconnection sites and generation of reactants for the resulting synthons. GraphRetro³⁰ identifies bond disconnection sites in the target molecule and then determines the leaving groups associated with the synthons. RetroPrime³⁵ employs dual Transformer models to separately address these tasks, incorporating atom mapping to enhance the interrelation between stages. Although semitemplate-based methods have shown promising results, the independent optimization of the two stages poses challenges for joint learning and increases computational complexity.

Template-free approaches utilize deep generative models to predict reactants without relying on predefined templates or leaving group libraries. These methods often employ sequence-to-sequence or graph-to-sequence architectures, representing molecules using the Simplified Molecular Input Line Entry System (SMILES).³⁶ The pioneering work by Schwaller et al. introduced the first Transformer-based sequence-to-sequence model for template-free retrosynthesis prediction, demonstrating that neural networks can learn chemical transformation patterns directly from reaction data without explicit template matching.³⁷ This approach was later extended with uncertainty calibration and attention-based interpretation, establishing the foundation for modern template-free retrosynthesis methods.³⁸ Building upon these foundations, Retroformer³⁹ combines molecular graph data with text-based representations, employs a local-global attention mechanism, and integrates reaction

center predictions to assist the Transformer decoder in reactant generation. UAlign⁴⁰ leverages graph attention networks to extract molecular features and aligns the generation process through a depth-first search of the target molecule. Template-free methods excel at enabling chemical reasoning over a broader reaction space; yet, their performance often varies significantly across different reaction classes.

Despite the progress in these methods, existing approaches often neglect the imbalance in reaction class distributions. As shown in Figure 1, we analyze the data distribution of the

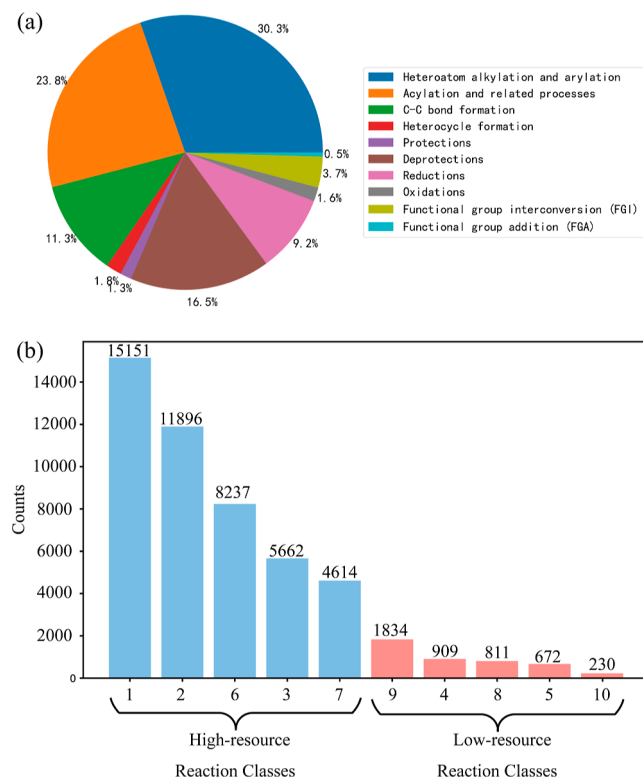


Figure 1. Statistical overview of reaction types in the USPTO-50K data set. (a) Classification and proportional distribution of the 10 reaction types. (b) Reaction classes ranked by descending counts of reactions.

USPTO-50K data set,⁴¹ a widely used benchmark for retrosynthesis tasks. Figure 1a displays the names and relative proportions of the 10 reaction classes in the USPTO-50K data set, while Figure 1b presents their respective data counts. The data set categorizes reactions into 10 distinct classes, with a marked disparity between high-resource classes (top-5 classes with the most samples) and low-resource classes (remaining 5 classes). This significant data imbalance poses challenges for model development. Efforts to improve performance on low-resource classes often result in degradation or forgetting in high-resource classes due to data limitations or distributional imbalances.

To address these challenges, we propose two novel distillation-based frameworks, Retro-MD and Retro-SD, that utilize mutual distillation^{42,43} and self-distillation^{44,45} strategies, respectively. Retro-MD employs distinct sampling temperatures to focus on high- and low-resource reaction classes, enabling the models to reinforce each other through mutual distillation during training. In contrast, Retro-SD extracts knowledge from previous high-performing iterations within the

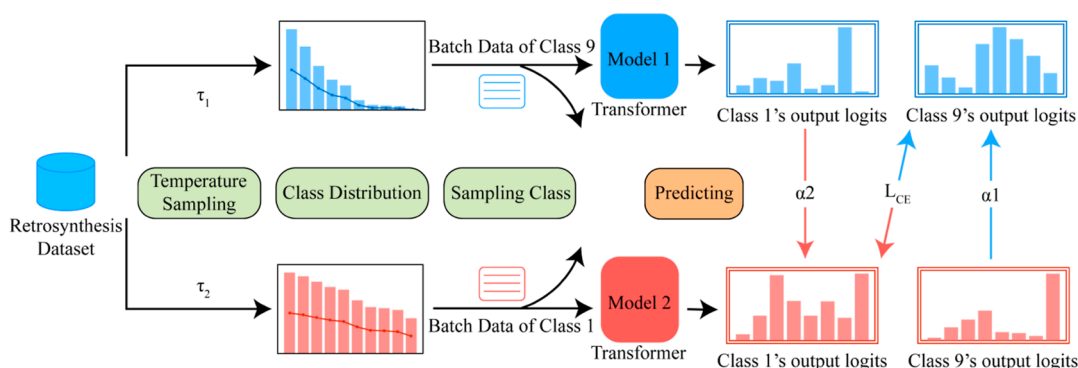


Figure 2. Overview of the Retro-MD framework. By utilizing temperature parameters τ_1 and τ_2 , Retro-MD generates distinct sampling distributions, each used to train specific reaction class batches. For instance, batches from Class 1 and Class 9 are processed alternately, with each model serving as both student and teacher during the training process. The distillation weights α_1 and α_2 dynamically adjust to determine the extent of learning from the peer model.

training process, treating these as teacher models to enhance the subsequent learning. Our experimental results demonstrate that these distillation strategies achieve state-of-the-art performance in template-free methods and competitive results in template-based and semitemplate-based methods.

2. RESULTS

2.1. Overview of the Single-Step Retrosynthesis Distillation Framework

Our proposed distillation strategy is model-agnostic and can be integrated with various architectures including graph neural networks and Transformers. In this study, we implement the strategy within Transformer models,⁴⁶ adopting a template-free approach to address the retrosynthesis prediction problem. Supporting Information Figure S1 illustrates the overall architecture of the Transformer model.

For data sets such as USPTO-50K,⁴¹ each reaction entry is categorized into a specific reaction class, enabling the formulation of distinct reaction class prediction tasks. However, data sets like USPTO-Full³² lack explicit reaction class annotations. To address this challenge, we employ MolCLR,⁴⁷ a pretrained model fine-tuned on product molecules and reaction classes from the USPTO-50K training, validation, and test sets. This fine-tuning enables the effective classification of reaction data in USPTO-Full. Although USPTO-Full includes a broader range of reaction classes than USPTO-50K, our classification methodology clusters these reactions into ten superclasses, providing a meaningful and manageable degree of partitioning for subsequent modeling.

During data set sampling, the training distribution is represented by a probability distribution P , which is defined in eq 1:

$$P(r) = \frac{M_r}{\sum_j M_j} \quad (1)$$

where M_r represents the number of samples in the training set D_{train} that belong to reaction class r , and $\sum_j M_j$ denotes the total number of reaction samples in the training set.

In conventional training, low-resource reaction classes exhibit lower sampling probabilities, often leading to underrepresentation and degraded performance for these classes. To mitigate this issue, we adopt a temperature sampling

technique,⁴⁸ rescaling the training distribution with a temperature hyperparameter τ , as shown in eq 2:

$$P(r) = \frac{M_r^{1/\tau}}{\sum_j M_j^{1/\tau}} \quad (2)$$

where τ is the temperature parameter. When $\tau > 1$, the sampling distribution becomes smoother, increasing the relative sampling probabilities of low-resource reaction classes. This adjustment amplifies the model's attention to underrepresented classes, effectively addressing the data imbalance and mitigating the associated biases. By controlling the value of τ , we achieve a balance between retaining the representation of high-resource classes and enhancing the focus on low-resource classes, thereby improving the overall model performance.

As shown in Figure 2, the Retro-MD framework involves the simultaneous training of two Transformer models, referred to as Model 1 and Model 2.⁴⁶ Each Transformer consists of an encoder and a decoder. To enhance the efficacy of mutual distillation while maintaining diversity between the models, we adjusted their sampling distributions for reaction classes. Specifically, Model 1's sampling distribution is controlled by a temperature parameter τ_1 , which biases the model toward high-resource reaction classes, whereas Model 2 uses a smoother distribution governed by τ_2 , increasing the representation of low-resource classes.

Initially, both models undergo standard training to build a robust understanding of retrosynthetic prediction and prevent premature or uninformative distillation. Once initialized, reaction classes are sampled from Model 1's distribution to determine the class r for training. Corresponding batch data from D_{train} are extracted and processed by both models. During this phase, Model 1 acts as the student, learning from Model 2, which serves as the teacher. This process is repeated with reaction classes sampled from Model 2's distribution, with the roles of teacher and student reversed.

After training on a batch sampled from one distribution, the next batch is immediately sampled from the other distribution. This alternating process ensures that the two models iteratively act as both a teacher and student, fostering mutual reinforcement. At the end of each epoch, the distillation weights α_1 and α_2 are dynamically updated to better balance the reliance on the peer model's output, thus enhancing the overall performance of both models.

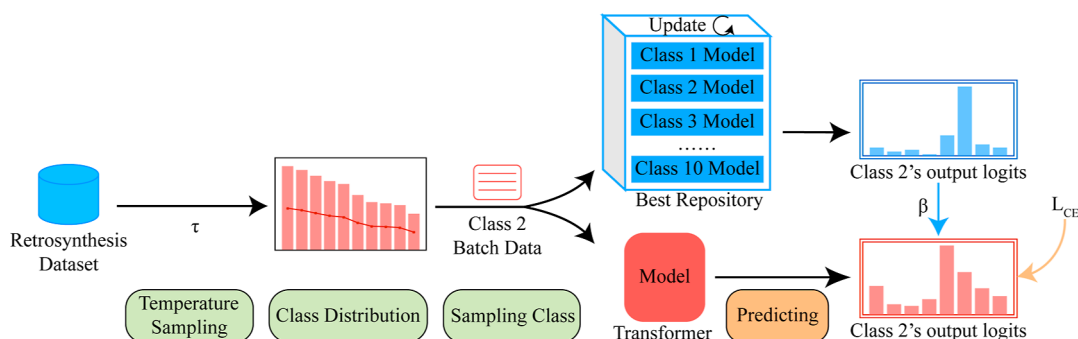


Figure 3. Overview of the Retro-SD framework. Using the temperature parameter τ , Retro-SD generates sampling distributions to determine the reaction class batch for training, as exemplified by Class2. The batch data for Class2 is processed by both the current model and the corresponding best-performing model from the repository to obtain predictions. The current model then computes its loss based on both the cross-entropy loss and the knowledge distillation loss, with the distillation weight β representing the extent of reliance on the teacher model's outputs. By iteratively updating the repository and refining the model's predictions through self-distillation, Retro-SD ensures that the current model benefits from the knowledge of historically optimal models, thereby enhancing the overall training process and predictive performance.

As illustrated in Figure 3, Retro-SD employs a single Transformer model throughout the entire training process, consisting of an encoder and a decoder. To facilitate self-distillation, a best model repository is established to store the checkpoint indices and loss values of the top-performing models for each reaction class prediction task. Initially, all loss values in the repository are set to infinity.

The training process begins with sampling a reaction class r from the training distribution, governed by a temperature parameter τ . The corresponding batch data for r is then extracted from the training set D_{train} . If no best model exists for the selected reaction class in the repository, then the current model computes only the cross-entropy loss. Otherwise, the current model performs distillation learning by leveraging the knowledge source from the repository's best-performing model for that reaction class. Here, we define the source model (best-performing model) parameters or prediction outputs as the "knowledge source", and the current model learns to imitate this through loss function design (e.g., KL divergence between the prediction distributions).

At the end of each epoch, the model's performance is validated using the validation set, and the loss values for each reaction class are recorded. If the loss value for a particular reaction class is lower than the corresponding value in the best model repository, the repository is updated with the current model's checkpoint index and loss value. By continuously updating the repository, Retro-SD dynamically leverages the most effective historical models to guide the current model's training, achieving an iterative process of self-distillation.

2.2. Baselines

We compare the performance of our models with state-of-the-art (SOTA) single-step retrosynthesis prediction methods across three categories: template-based, semitemplate-based, and template-free methods. For the template-based category, we select RetroSim,³¹ GLN,³² LocalRetro,³⁴ and RetroKNN.⁴⁹ Among the semitemplate-based approaches, we include GraphRetro,³⁰ RetroPrime,³⁵ and RetroExplainer.⁵⁰ For template-free methods, we employ Transformer,⁴⁶ SCROP,⁵¹ MEGAN,⁵² Retroformer,³⁹ and UAlign⁴⁰ as baselines. For the baseline models used in our comparisons, we adopt the experimental results reported in their respective publications, ensuring consistency in parameter configurations and settings. Supporting Information Tables S1 and S2 provide detailed parameter settings for Retro-MD and Retro-SD, respectively.

2.3. Top-N Accuracy

Table 1 summarizes the performance of Retro-MD and Retro-SD, along with baseline models, on the USPTO-50K data set

Table 1. Top- k Exact Match Accuracy(%) for Retrosynthesis Prediction on the USPTO-50K Data Set^a

model	top-1	top-3	top-5	top-10
template-based methods				
RetroSim	52.9	73.8	81.2	88.1
GLN	64.2	79.1	85.2	90.0
LocalRetro	63.9	86.8	92.4	96.3
RetroKNN	66.7	88.2	93.6	96.6
semitemplate-based methods				
GraphRetro	63.9	81.5	85.2	88.1
RetroPrime	64.8	81.6	85.0	86.9
Retroexplainer	66.8	88.0	92.5	95.8
template-free methods				
Transformer	57.1	71.5	75	77.7
SCROP	59.0	74.8	78.1	81.1
MEGAN	60.7	82.0	87.5	91.6
Retroformer	64.0	82.5	86.7	90.2
UAlign	66.4	86.7	91.5	95.0
Retro-MD ₁ (ours)	67.8	87.5	91.6	94.5
Retro-MD ₂ (ours)	67.5	87.1	91.6	94.2
Retro-SD(ours)	68.9	87.9	91.9	94.6

^aBest performance is in bold.

with reaction class known. We denote the two Retro-MD variants trained under different sampling temperatures as Retro-MD₁ and Retro-MD₂. Notably, Retro-MD₁ achieves 67.8% top-1 and 94.5% top-10 accuracy, while Retro-SD attains 68.9% top-1 and 94.6% top-10 accuracy. Our proposed methods set new state-of-the-art performance among template-free approaches and are competitive with leading template-based and semitemplate-based methods. Both RetroKNN and RetroExplainer incorporate additional molecular graph information, resulting in higher accuracy. In particular, Retro-MD₁ surpasses Retroformer by 3.6% in top-1 accuracy and 4.3% in top-10 accuracy, while Retro-SD exceeds Retroformer by 4.9% and 4.4% in top-1 and top-10 accuracy, respectively. Visualization results of a randomly selected test example from the USPTO-50K data set for Retro-SD and Retroformer are shown in Supporting Information Figure S4. The Top-K

accuracy for each reaction category on the USPTO-50K data set is reported in Table S6. Although the temperature parameter results in slight variations in resource performance, its key contribution lies in supporting complementary sampling distributions, which facilitate effective mutual distillation. The performance improvement does not stem from the temperature scaling itself but from the collaborative knowledge transfer between the models it supports.

Table 2 summarizes the top-*k* exact match accuracy for retrosynthesis prediction on the USPTO-Full data set. To

Table 2. Top-*k* Exact Match Accuracy(%) for Retrosynthesis Prediction on the USPTO-Full Data Set^a

model	top-1	top-3	top-5	top-10
template-based methods				
RetroSim	32.8	-	-	56.1
GLN	39.3	-	-	63.7
LocalRetro	39.1	53.3	58.4	63.7
semitemplate-based methods				
RetroPrime	44.1	59.1	62.8	68.5
RetroExplainer	51.4	70.7	74.7	79.2
template-free methods				
MEGAN	33.6	-	-	63.9
Retro-MD ₁ (ours)	42.7	60.3	65.7	70.6
Retro-MD ₂ (ours)	42.2	59.9	65.7	70.7
Retro-SD(ours)	42.3	60.5	66.4	71.8

^aMissing accuracy values are denoted by “-”.

preliminarily assess the distillation strategy's efficacy on unseen reaction types under computational constraints, we restrict our comparisons to models evaluated on the USPTO-Full data set, as reported in prior works. We employ a classification model to adapt our approach for this larger data set. Experimental results show that Retro-MD₁ achieves top-1 and top-10 accuracies of 42.7% and 70.6%, respectively, surpassing the best template-based method by 3.6% and 6.9%. Retro-SD obtains top-1 and top-10 accuracies of 42.3% and 71.8%, exceeding the best template-based method by 3.2% and 8.1%. Both models exhibit strong performance on the USPTO-Full data set, although they underperform relative to RetroExplainer, a semitemplate-based method. Our methods surpass current template-based approaches, indicating that as the data set grows, template-based methods struggle to handle novel reactions outside their template libraries, which in turn diminishes predictive performance. For semitemplate-based approaches, RetroExplainer employs a graph Transformer that leverages additional molecular graph information, including bond embeddings, to achieve higher accuracy. Nevertheless, by relying solely on SMILES representations, our models can still make predictions on unclassified data through a pretrained classification model and achieve competitive performance.

2.4. Visualization and Reactant Generation

To evaluate the predictive performance of our proposed models compared to baseline methods, we present in Figure 4 the reactant prediction results for Retro-MD₁, Retro-MD₂, Retro-SD, and Retroformer on a representative example (US20050019696A1). Notably, while none of the models identified the ground-truth reactants in their Top-1 predictions, our models successfully retrieved the correct reactants in their Top-2 results, outperforming Retroformer, which required Top-3 predictions to achieve the same result. This

demonstrates our models' superior ability to prioritize correct reactants. Furthermore, Retroformer exhibited limited prediction diversity, generating identical molecules for both the Top-1 and Top-5 positions.

Figure 5 details our model's output processing pipeline, applicable to both trained Retro-MD and Retro-SD variants. The implementation of root-alignment-based data augmentation and prediction reranking significantly enhances both the diversity and chemical validity of the generated predictions.

2.5. Few-Shot Study

To explore the domain adaptation capabilities of our proposed methods, we converted the USPTO-50K data set into a few-shot version. Our data processing approach follows that of Retro-KNN.⁴⁹ Specifically, we reduced the data for reaction classes 6–10 to 10% of each class's reactions while leaving other classes unchanged and preserving the original relative order. We then compared Retro-MD and Retro-SD with the Retroformer baseline to assess performance on this modified data set.

Table 3 presents the test results for Retro-MD, Retro-SD, and Retroformer on reaction classes 6–10 under a few-shot setting. Our methods yield more accurate reactant predictions overall. Retro-MD₂ achieves top-1 and top-10 accuracies of 57.1% and 89.8%, respectively, exceeding Retroformer by 10.4% and 10.0%. Meanwhile, Retro-SD attains top-1 and top-10 accuracies of 57.0% and 87.4%, outperforming Retroformer by 10.3% and 7.6%. These results demonstrate that even with limited data, our approaches significantly enhance predictive performance, a capability that will likely become increasingly important in this field.

Figure 6 displays the top-5 prediction results of Retro-MD₁, Retro-MD₂, Retro-SD, and Retroformer for a randomly selected example with reaction class 6 in the test set. All four models successfully ranked the correct reactants at the Top-1 position. Figure 7 presents the top-5 predictions for a randomly chosen reaction class 7 example, where Retroformer generated an invalid SMILES in its fourth prediction—further validating the superior performance of our models. Extended validation results for reaction classes 8–10 are provided in Supporting Information Figures S3–S5, respectively. Collectively, our models consistently guarantee both chemical validity and novelty across all predictions.

2.6. Ablation Study

We also performed ablation experiments on the USPTO-50K data set to evaluate how our classification approach and distillation strategy influence retrosynthesis prediction. As shown in Table 4, Retro-MD_{1,2}^c outperform Retro-MD_{1,2}^a and Retro-MD_{1,2}^b across all reaction classes, including both high-resource and low-resource classes, underscoring the importance of classifying data by reaction class and applying the mutual distillation strategy. Specifically, configuration Retro-MD_{1,2}^c demonstrates superior performance compared to Retro-MD_{1,2}^a, indicating that mutual distillation can further improve outcomes when data are grouped. Additionally, Retro-MD_{1,2}^c also surpasses Retro-MD_{1,2}^b, suggesting that randomly splitting the training data introduces substantial confusion across different retrosynthesis tasks, thereby disrupting the update of distillation weights and leading to performance degradation. These findings indirectly validate the rationale and effectiveness of categorizing retrosynthesis predictions by reaction class within the distillation learning process.

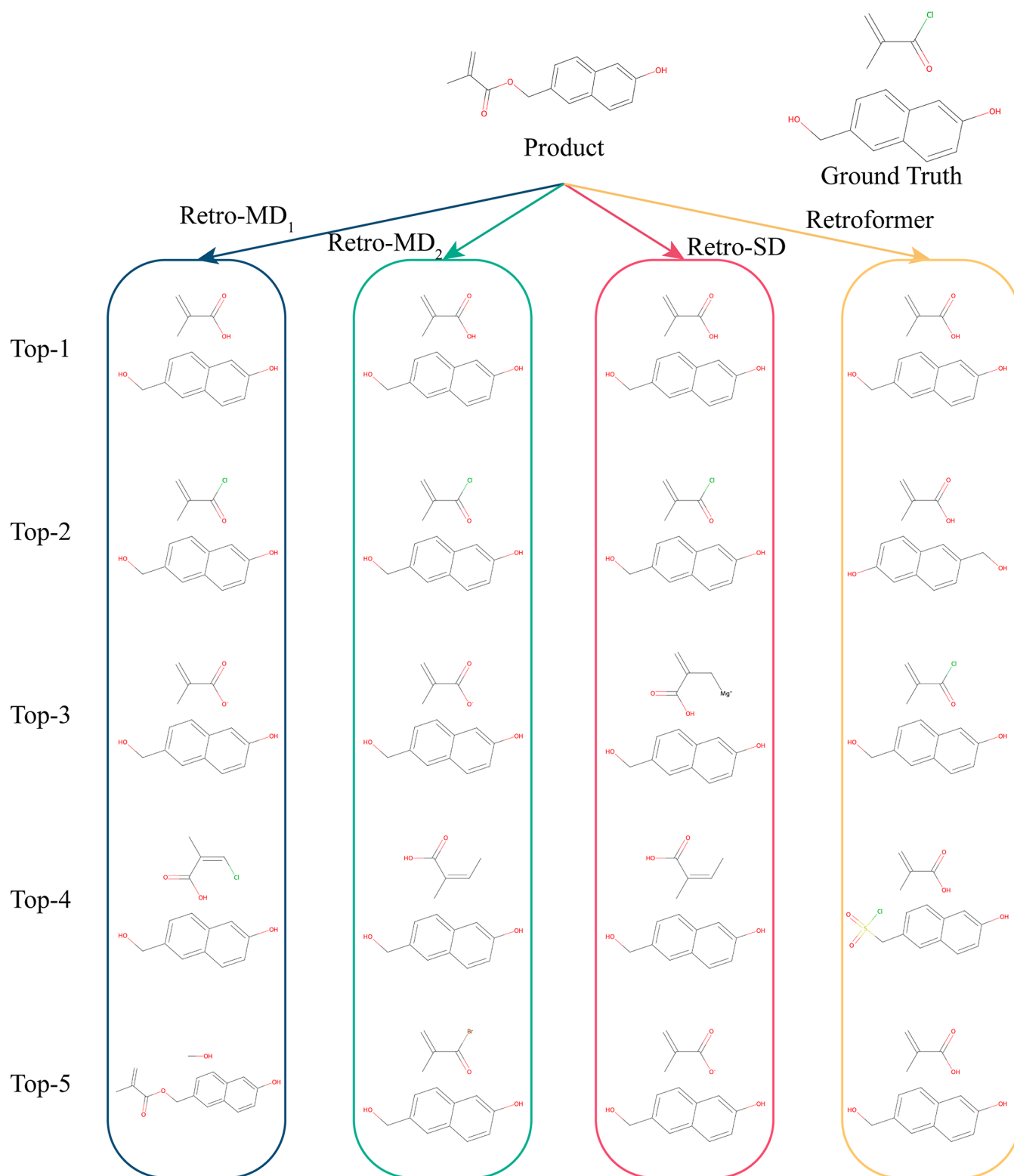


Figure 4. Randomly selected example (US20050019696A1) with reaction type 2. In this example, Retro-MD₁, Retro-MD₂, and Retro-SD all correctly predicted the reactants within the Top-2 results, while Retroformer achieved the correct prediction at Top-3.

Under the Retro-SD method, for overall classes as well as high-resource and low-resource reaction classes, Retro-SD^e outperforms Retro-SD^a in both top-1 and top-10 accuracy. This result demonstrates the strong effectiveness of self-distillation under a well-designed data classification scheme. Retro-SD^e outperforms Retro-SD^d across all retrosynthesis

prediction metrics, corroborating our earlier conclusion that precise task classification facilitates more effective distillation learning, whereas conflating tasks impedes standard model training. In addition, Supporting Information Tables S6 and S7 present the experimental results of Retro-MD and Retro-SD under different temperature hyperparameters. The results

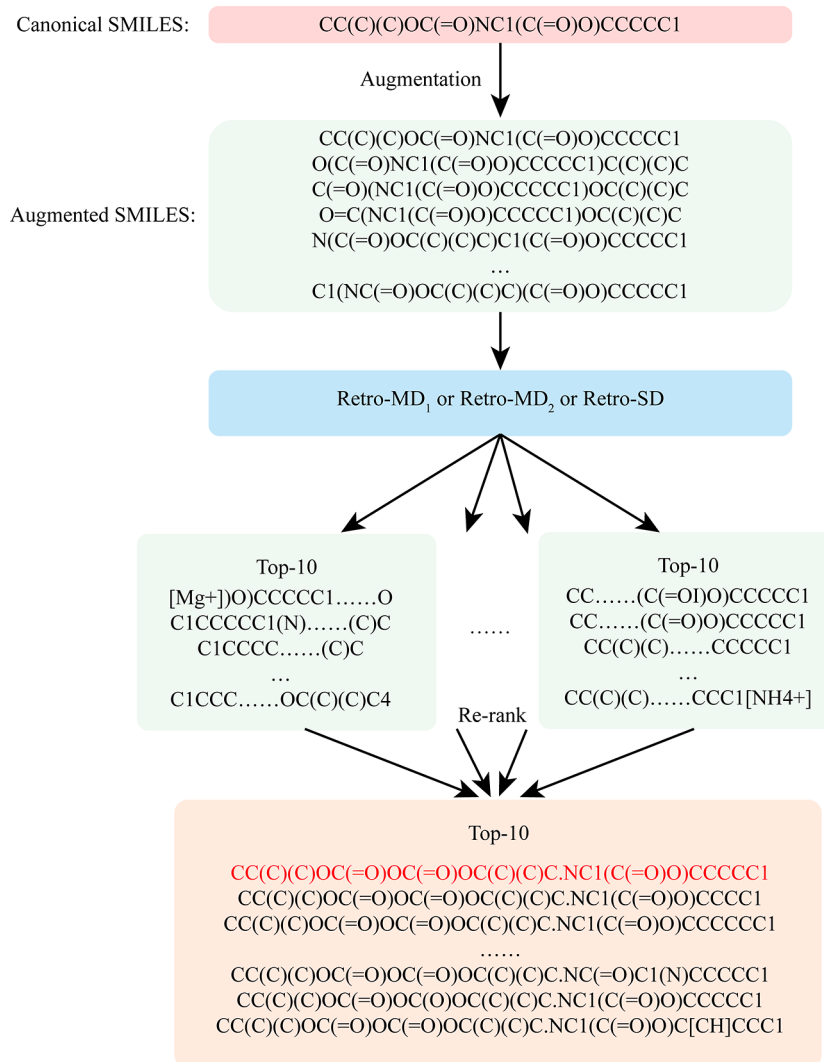


Figure 5. Demonstration of trained models performing reactant sequence prediction. The workflow shows: (1) target molecule processing, (2) model prediction generation, and (3) chemical validity-based reranking (by RDKit). Chemical entities including ions (e.g., [Mg⁺]) appear as they are included in our extracted dictionary (see Table S5 for dictionary details).

Table 3. Top-*k* Exact Match Accuracy (%) on Reaction Classes 6–10 of the USPTO-50K Data Set under Few-Shot Settings

model	top-1	top-3	top-5	top-10
Retroformer	46.7	69.1	75.4	79.8
Retro-MD ₁ (ours)	56.9	79.8	83.8	88.0
Retro-MD ₂ (ours)	57.1	80.8	85.5	89.8
Retro-SD(ours)	57.0	78.4	82.7	87.4

indicate that the temperature has little effect on Retro-MD, whereas the performance of Retro-SD decreases as the temperature increases. We hypothesize that this is due to the negative impact of excessively resampled low-resource data on the model.

In summary, the mutual distillation strategy trains two models on distinct reaction class sampling distributions, which may encourage each model to explore different directions in the parameter space. This dual exploration could potentially facilitate the convergence of effective parameters during the distillation process. On the other hand, the self-distillation strategy appears to focus on parameter spaces that are

particularly effective for specific reaction classes. By learning soft labels from earlier models, the current model may further refine its overall performance in retrosynthesis prediction.

3. DISCUSSION

In this study, we propose Retro-MD and Retro-SD, two methods that employ mutual distillation and self-distillation strategies, respectively, to enhance the retrosynthesis prediction performance. Retro-MD maintains prediction diversity by simultaneously training two models with different sampling temperatures, thereby achieving performance improvements across both high- and low-resource reaction classes. Retro-SD trains a single model that continuously integrates the knowledge from checkpoints that perform optimally on individual reaction classes during training, achieving a superior predictive performance.

Experimental results demonstrate that on the USPTO-50K data set, Retro-MD and Retro-SD outperform all other single-step retrosynthesis methods in Top-1 accuracy and maintain competitive performance with the best methods in Top-3 and Top-5 metrics, and although slightly inferior to the optimal model in Top-10, they still significantly outperform template-

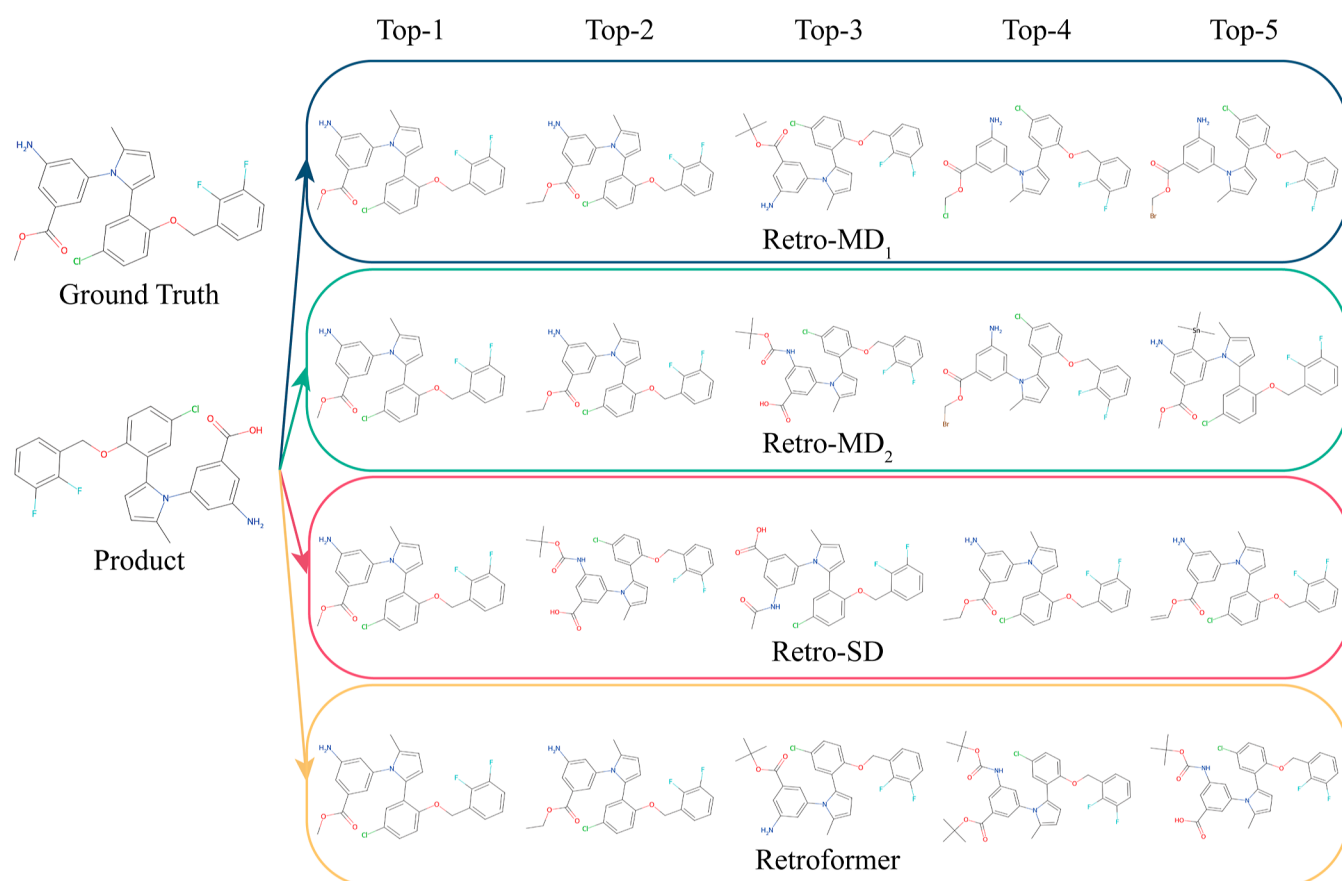


Figure 6. Randomly selected example (US20070082912A1) with reaction type 6. Retro-MD₁, Retro-MD₂, Retro-SD, and Retroformer all successfully identified the correct reactants as their top-ranked (Top-1) predictions.

free methods such as Retroformer³⁹ and Ualign.⁴⁰ For the USPTO-Full data set, which contains unknown reaction classes, we propose a retrosynthesis classification framework to address this challenge. This framework enhances the model's attention to low-resource classes through reaction data classification. Experimental results show that both Retro-MD and Retro-SD outperform all template-based methods as well as template-free methods like Retroformer and Ualign, though they perform slightly worse than semitemplate-based methods.

In the experimental setting of few shots, we reduced the training sets of reaction classes six through ten to one-tenth of their original size to simulate realistic scenarios of a few shots.⁴⁹ These reaction classes already constitute a smaller proportion of the entire data set, making this setting more representative of few-shot conditions after the reduction. Due to the unavailability of specific data for Retro-KNN,⁴⁹ we compared our methods with Retroformer. The results show that our methods achieve an improvement of 10.2%–10.4% over Retroformer in Top-1 accuracy and 7.6%–10.0% in Top-10 accuracy, confirming the superiority of our approach in the scenario of few shots.

Mechanistic analysis reveals that the USPTO-50K data set exhibits significant variations in data volume across reaction classes, different levels of understanding difficulty, and complex interclass relationships, leading to substantial differences in prediction accuracy across different reaction classes on the test set. For the USPTO-Full data set, since the classification model is trained on USPTO-50K, it also contains other reaction types

not covered in the ten reaction classes, such as dehalogenation reactions. To address these challenges, both Retro-MD and Retro-SD employ distillation strategies. Retro-MD adjusts model training preferences through oversampling of low-resource reaction classes to improve performance in these classes, while mutual distillation ensures that prediction accuracy on high-resource reaction classes remains unaffected. Retro-SD draws inspiration from experiences in natural language processing with multilingual translation tasks, recognizing that in multiobjective optimization problems, individual prediction accuracies cannot simultaneously achieve optimality. This method enables the model to capture optimal parameters for each reaction class during training and perform distillation learning through a self-distillation strategy, improving the prediction accuracy across all reaction types without altering the original data distribution. Our experimental observations from Table S6 suggest that mutual distillation enables the models to learn complementary aspects of the reaction space. However, we acknowledge that a definitive mechanistic understanding would require more extensive investigation beyond the scope of this work.

Recent advances in data set quality assurance, exemplified by AutoTemplate,⁵³ have established rigorous systematic protocols for rectifying atom-mapping errors, recovering missing reactants, and eliminating inconsistent reactions in chemical reaction databases. This work demonstrates the critical importance of high-quality curated data for developing reliable machine learning models in retrosynthesis. Similarly, multi-stage machine learning frameworks⁵⁴ that integrate classifica-

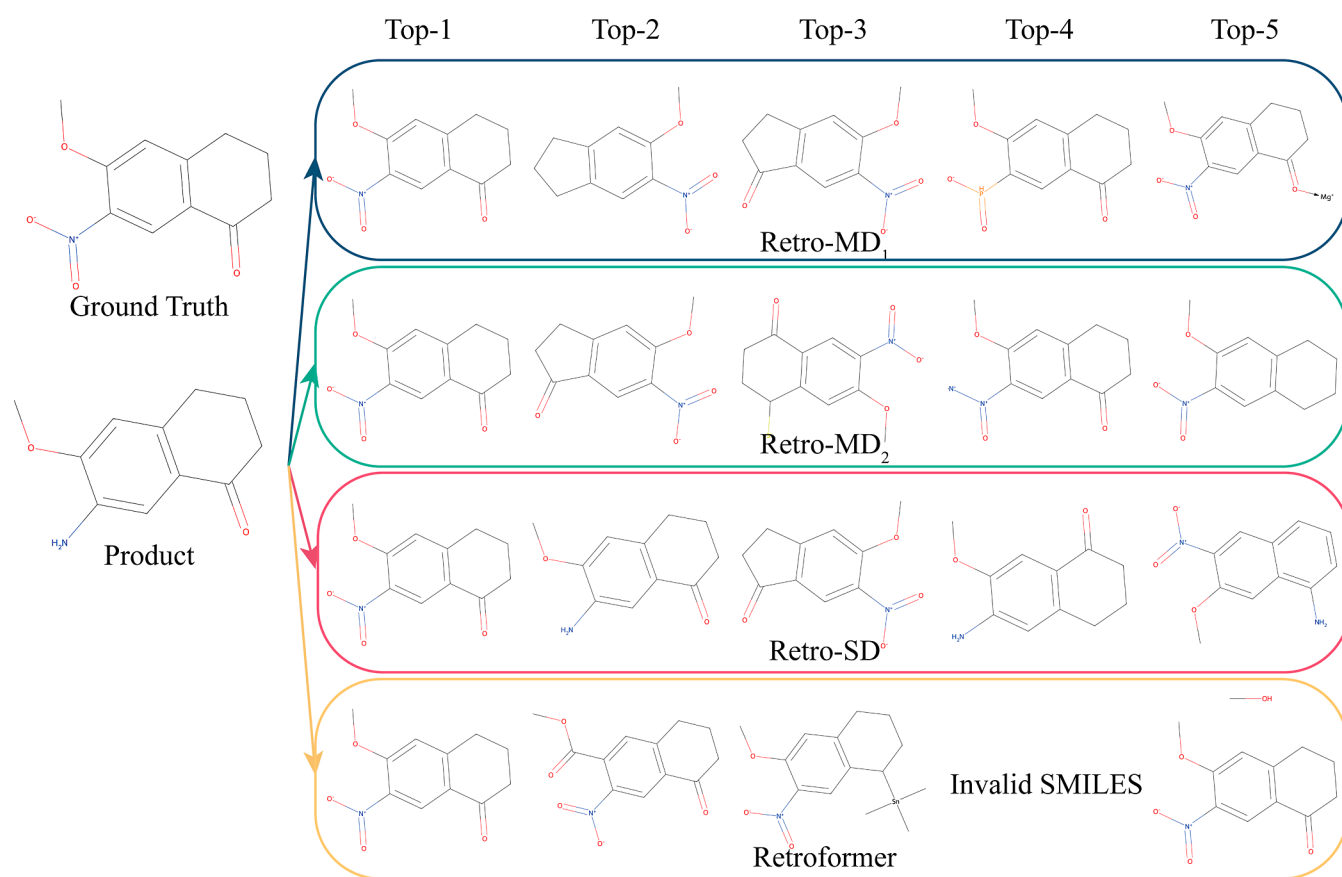


Figure 7. Randomly selected example (US20020055643A1) with reaction type 7. All four models successfully identified the ground-truth reactants in their Top-1 predictions, whereas Retroformer produced a chemically invalid structure at the Top-4 position.

Table 4. Ablation Study on the Effects of Data Classification and Knowledge Distillation^a

method	overall		high-resource		low-resource	
	top-1	top-10	top-1	top-10	top-1	top-10
retro-MD ₁ ⁱ	67.0	93.0	66.5	93.0	72.8	92.0
retro-MD ₂ ⁱ	66.2	93.6	65.7	93.7	70.8	92.0
retro-MD ₁ ^b	53.3	86.6	53.1	86.6	55.8	86.4
retro-MD ₂ ^b	51.4	85.6	51.4	85.7	51.6	85.0
retro-MD ₁ ^c	68.0	94.2	67.3	94.3	75.0	93.1
retro-MD ₂ ^c	66.8	93.7	66.1	93.8	74.1	93.1
retro-SD ^a	67.9	93.4	67.1	93.6	76.1	91.5
retro-SD ^d	52.4	87.2	52.2	87.3	54.7	86.6
retro-SD ^e	68.9	95.1	68.2	95.3	76.1	93.5

^aFor Retro-MD, predictive accuracies of Model 1 and Model 2 are reported with τ set to 1 and 5; for Retro-SD, τ is set to 5. Configurations are denoted as follows: ⁱOrdered Data, ^bShuffled Data + Mutual Distillation, ^cOrdered Data + Mutual Distillation, ^dShuffled Data + Self Distillation, ^eOrdered Data + Self Distillation. "Ordered Data" indicates partitioning by original reaction classes; "Shuffled Data" refers to random partitioning.

tion and ranking modules for synthesis-related predictions represent a significant advancement in computer-aided synthesis planning (CASP), enabling more sophisticated and accurate prediction pipelines. Our distillation approach complements these developments by providing a training methodology that is inherently robust to data quality variations through its ensemble-like mutual distillation mechanism. We believe that the integration of rigorous data curation protocols

such as AutoTemplate with robust training frameworks like ours represents a synergistic direction for enhancing the reliability and generalizability of chemical reaction prediction models.

The limitations of this study are manifested primarily in the following aspects: first, we only use the Transformer architecture within the fairseq framework to evaluate the effectiveness of distillation strategies, which could be extended to more advanced model architectures in the future. Second, our method requires data preclassification prior to training execution, which limits its applicability in out-of-distribution scenarios. However, it is important to note that our approach incorporates a preclassification model that does not require high-quality reaction class annotations and can address OOD (Out-of-Distribution) issues, which goes beyond template-based methods. The preclassification model serves as a flexible framework that can adapt to new reaction types by learning from the chemical features themselves, rather than relying on predefined templates. This capability makes our method more robust to OOD scenarios compared to traditional template-based approaches, which are constrained by their template libraries. Additionally, we employ only root-alignment-based data-augmentation methods to obtain alignment information between atoms in reactants and products, and future work could integrate more chemical knowledge to guide model training. Finally, the variability of tautomers represents an additional chemical consideration beyond the standard SMILES representation. Inconsistent forms of tautomers in the training data may affect model performance in the following ways: (1) causing significant mismatches between

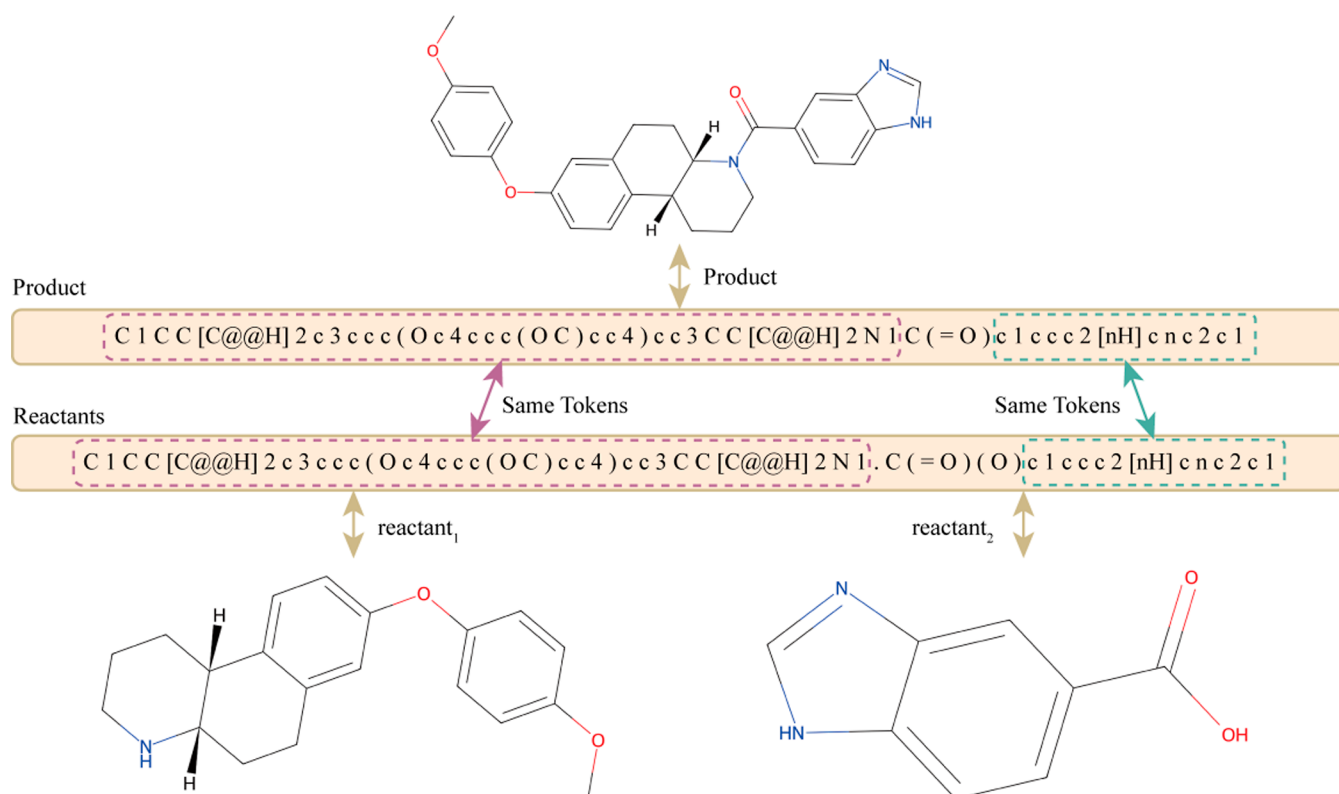


Figure 8. Randomly selected examples after root-aligned data augmentation. The top row shows the SMILES and image representation of the product, while the bottom row shows the SMILES and image representations of the corresponding list of reactants. It can be observed that most tokens in the product SMILES have one-to-one correspondences with those in the reactant SMILES, which significantly reduces the training difficulty and enhances the model's ability to capture local molecular structures.

chemically equivalent structures; (2) confusing the model with multiple valid representations of the same molecule; and (3) limiting the generalization ability of reactions involving tautomeric equilibria. Furthermore, during the data augmentation process, the model may fail to adequately learn the molecular consistency between tautomers, which could have a potential impact on its final performance. In conclusion, our proposed Retro-MD and Retro-SD methods effectively improve single-step retrosynthesis prediction performance in few-shot scenarios through innovative distillation strategies, providing new insights and methods for addressing prediction challenges in low-resource reaction classes.

4. CONCLUSIONS

In conclusion, Retro-MD and Retro-SD significantly enhance single-step retrosynthesis prediction through innovative mutual and self-distillation strategies. Our methods achieve state-of-the-art Top-1 accuracy on the USPTO-50K data set and demonstrate strong performance in both few-shot and full-data scenarios, particularly for low-resource reaction classes. While slightly inferior in Top-10 accuracy to some semitemplate-based approaches, they consistently outperform existing template-free and template-based baselines. The proposed framework effectively addresses data imbalance, offering a robust solution for practical retrosynthesis planning.

5. METHODS

5.1. Training Loss

The distillation process relies on minimizing the divergence between the output probability distributions of the teacher and student models. The knowledge distillation loss is defined in eq 3:

$$L_{\text{KD}}(\theta_S, \theta_T) = - \sum_{t=1}^T \sum_{i=1}^V p(y_{t,i}^T | x; \theta_T) \cdot \log p(y_{t,i}^S | x; \theta_S) \quad (3)$$

where θ_T and θ_S represent the parameters of the teacher and student models, respectively. Here, T represents the effective length of the ground truth sequence, x is the SMILES sequence of the target molecule, V is the vocabulary comprising various tokens, $y_{t,i}^T$ is the label predicted by the teacher model at position t for the i -th sample, and $y_{t,i}^S$ is the label predicted by the student model at position t for the i -th sample.

Simultaneously, each model computes a cross-entropy loss for the batches sampled from its own distribution, as given in eq 4:

$$L_{\text{CE}}(\theta_S) = - \sum_{t=1}^T \sum_{i=1}^V y_{t,i} \log(y_{t,i}^S | x; \theta_S) \quad (4)$$

where $y_{t,i}$ is the true label at position t for the i -th sample.

For the student model, the overall loss function combines the cross-entropy loss and the knowledge distillation loss, weighted by the distillation coefficient $\alpha_S[r]$, as formulated in eq 5:

$$L_{\text{MD}} = (1 - \alpha_S[r]) \cdot L_{\text{CE}}(\theta_S) + \alpha_S[r] \cdot L_{\text{KD}}(\theta_S, \theta_T) \quad (5)$$

where $\alpha_S[r] \in (0, 1)$ indicates the degree of reliance on the teacher model's predictions for reaction class r .

The total loss function for Retro-SD is defined in eq 6:

$$L_{\text{SD}} = L_{\text{CE}}(\theta_S) + \beta \cdot L_{\text{KD}}(\theta_S, \theta_T) \quad (6)$$

where β is a hyperparameter that controls the weight of the distillation loss, determining the degree to which the student model relies on the teacher model's predictions.

5.2. Data Preparation

To assess the robustness of Retro-MD and Retro-SD, we used two widely recognized benchmark data sets, USPTO-50K⁴¹ and USPTO-Full.³² The USPTO-50K data set comprises 50,016 reaction entries, each specifying the reactants and products, and is organized into 10 distinct reaction classes. In contrast, the USPTO-Full data set is substantially larger, containing 1,013,118 reaction records. We adopted the same data splitting approach as previous researchers for all data sets, dividing the USPTO-50K data set and the USPTO-Full data set into 80%/10%/10% proportions. The data cleaning and splitting methods are the same as those in GLN.

5.2.1. Root-Aligned SMILES. We adopt the same processing method as R-SMILES.⁵⁵ We initially focused on decomposing SMILES into meaningful tokens following the regular expression approach proposed by Schwaller et al.³⁷ Given that common chemical substructures exist between reactants and target products in chemical reactions, prior studies have utilized atom mapping or substructure matching algorithms to identify these features. In this work, we employ atom mapping to detect such substructures. For single-step molecular retrosynthesis prediction, we randomly select an atom from the target product and designate it as the starting point to sequentially traverse the remaining atoms, thereby generating the product SMILES. Based on the order of atoms in the product, we identify the corresponding atoms in the reaction and arrange them as closely as possible to the respective sequence. Subsequently, all atom mappings are removed from the SMILES representations of both the target product and the reactants, yielding root-aligned data. The root-aligned SMILES of the target product and reactants exhibit a high degree of similarity, which aids the model in reducing the search space and enhancing the cross-attention mechanism. The random selected examples of product and reactant SMILES after the root alignment strategy are shown in Figure 8.

5.2.2. Data Augmentation. Following the root-aligned data processing approach, we applied 20-fold data augmentation to the training, validation, and test sets of the USPTO-50K data set and 5-fold data augmentation to the training, validation, and test sets of the USPTO-Full data set. As shown in Supporting Information Table S3, the splitting of the USPTO-50K and USPTO-Full data sets is consistent with R-SMILES.⁵⁵ Supporting Information Table S4 presents the number of samples in each reaction category after data augmentation for both data sets. During the data augmentation process, multiple augmented data entries were generated for the same reaction by selecting different starting atoms, thereby facilitating model training. Supporting Information Figure S2 illustrates how augmented data are fed into the model to generate multiple ranked prediction results. These predictions for the same product are then reranked to obtain the final ordered predictions.

5.3. Experimental Environment and Training Time

All experiments were conducted on a single NVIDIA RTX 4090 GPU (24GB) with 90GB of memory using PyTorch 1.10.0 and CUDA 11.3. Retro-MD training required approximately 23 h for 30 epochs, while Retro-SD completed 240 epochs in 40 h.

5.4. Evaluation Metrics

We evaluate retrosynthesis predictions using the top- k exact match accuracy metric. Specifically, we convert the top- k predicted sequences into standardized SMILES representations with RDKit.⁵⁶ If any of the predicted sets of reactants exactly matches the ground-truth set, we consider the prediction successful. The top- k exact match accuracy is computed as shown in eq 7:

$$\text{Exact Match}(k) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{Match}_i \in \sum_{j=1}^k \text{Predictions}_{i,j}) \quad (7)$$

where N is the number of samples in the test set D_{test} , $\mathbb{I}(\cdot)$ is the indicator function (which equals 1 if the condition inside the

parentheses is true and 0 otherwise), Match_i represents the ground-truth set of reactants for the i -th sample, and $\text{Predictions}_{i,j}$ denotes the j -th predicted set of reactants for the i -th sample.

■ ASSOCIATED CONTENT

Data Availability Statement

The code for our Retro-MD workflow is available at <https://github.com/yuzhou2333/Retro-MD>,⁵⁷ and the code for our Retro-SD workflow is available at <https://github.com/yuzhou2333/Retro-SD>.⁵⁸ The uspto50k data set used for model training can be found in both repositories.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.6c00322>.

Description of data: model training architecture and experimental parameters, additional experimental examples, and supplementary experiments on different reaction classes (PDF)

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Author Contributions

[#]Y.L. and Z.Y. contributed equally to this work. Y.P.L. and Y.S.L. conceived the ideas and designed the necessary steps of

the workflow. Y.P.L. and Z.Y. drafted the manuscript. Z.Y. performed the method implementation, benchmarked the hyperparameters used in the workflow by evaluating through test sets, and finalized the workflow with optimized settings. J.Y.Z. and Y.J.C. implemented the comparison with other SOTA methods. S.W., L.Y.W., X.X.Z., and Y.Y.H. analyzed and refined the workflow. Y.S.L. supervised the research. All authors contributed to the discussions and provided feedback on the manuscript.

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Notes

The authors declare no competing financial interest.

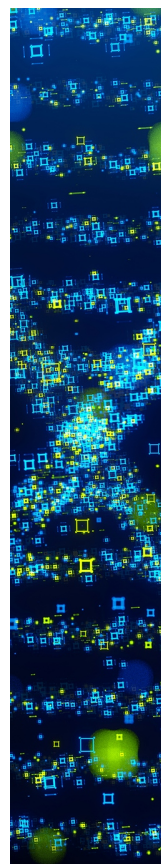
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